

Pipeline explicable para detección, segmentación y clasificación de lesiones mamarias: integración de radiomics y RAG sobre CBIS-DDSM

Reporte de Evidencia Empírica — Avance de Experimentos

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Programa: Maestría en Computación

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Institución: Universidad del Cauca

Dataset : CBIS-DDSM (Cancer Imaging Archive) – masas mamarias

Tarea 1 : Segmentación (U-Net / Attention U-Net + ResNet34/50)

Tarea 2 : Clasificación (EfficientNet-B0/B2/B3)

Tarea 3 : Fusión multimodal imagen + radiómicas (EfficientNet-B2 + PyRadiomics)

Este reporte presenta únicamente la evidencia empírica generada. No incluye conclusiones.

1. Segmentación de Masas Mamarias

U-Net / Attention U-Net • Encoder: ResNet34 / ResNet50 • Dataset: CBIS-DDSM masses

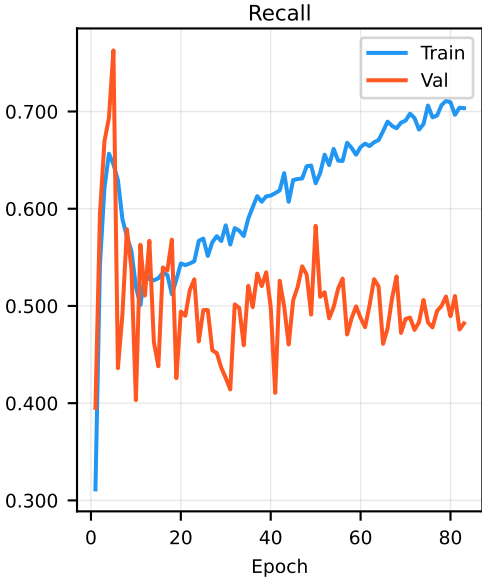
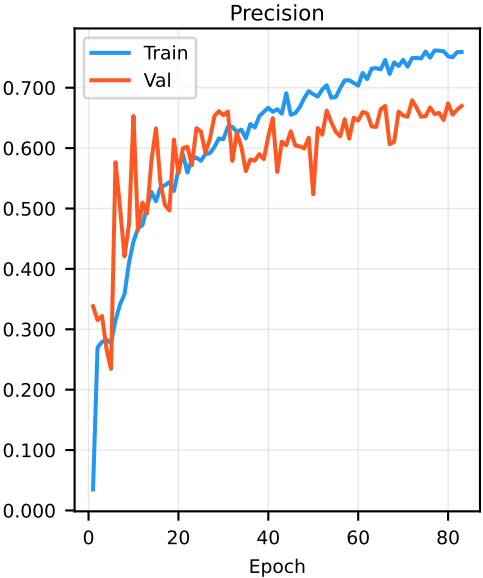
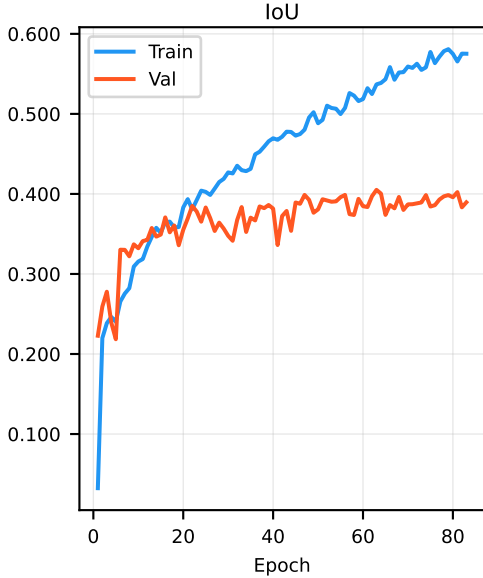
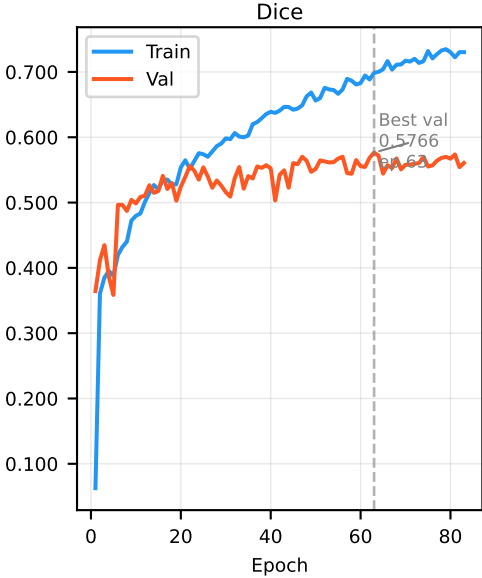
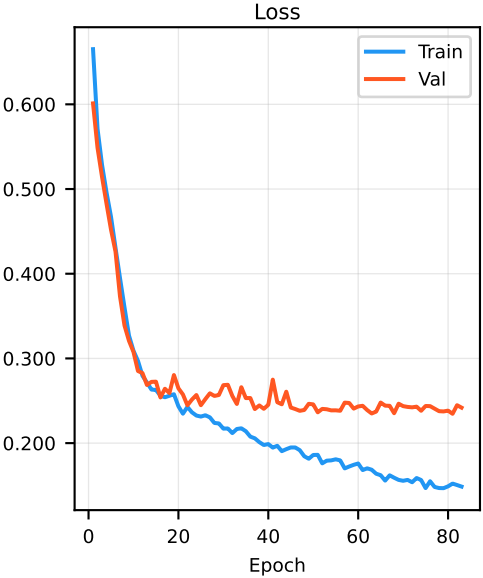
Resumen — Experimentos de Segmentación

Experimento	Arquitectura	Encoder	Patch	Loss	Best epoch	Best val Dice	Test Dice (media±std)
masses_experiment_01	unet	resnet34	full	-	63	0.5766	-
masses_exp02_resnet34_fo	unet	resnet34	full	focal_tversky	59	0.5772	-
masses_exp03_resnet50_ag	unet	resnet50	full	bce_dice	32	0.5451	-
masses_exp04_resnet34_fir	unet	resnet34	full	bce_dice	21	0.5423	-
masses_exp05_resnet50_fre	unet	resnet50	full	bce_dice	37	0.5314	-
masses_exp06_resnet34_fir	unet	resnet34	full	bce_dice	29	0.4706	-
masses_exp07_patch512_re	unet	resnet34	512	bce_dice	35	0.8347	0.8081
masses_exp08_attention_un	attention_unet	resnet34	512	bce_dice	25	0.8322	0.8125
masses_exp09_attention_un	attention_unet	resnet34	512	bce_dice	12	0.8158	-
masses_exp10_attunet_pat	attention_unet	resnet34	768	focal_tversky	12	0.7755	-
masses_exp11_attunet_pat	attention_unet	resnet34	768	bce_dice	49	0.7991	-
masses_exp12_attunet_pat	attention_unet	resnet34	512	bce_dice	60	0.8229	-

masses_experiment_01
U-Net + RESNET34 (pretrained ImageNet)

Configuración

Parámetro	Valor
Arquitectura	unet
Encoder	resnet34
Encoder weights	imagenet
In channels	1
Image size	1024
Batch size	-
Epochs	-
Aug profile	-
Optimizer	adamw
LR	-
Weight decay	-
Scheduler	-
T_max	-



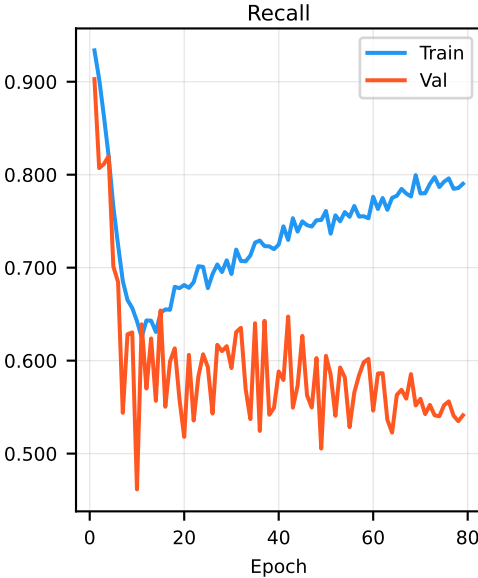
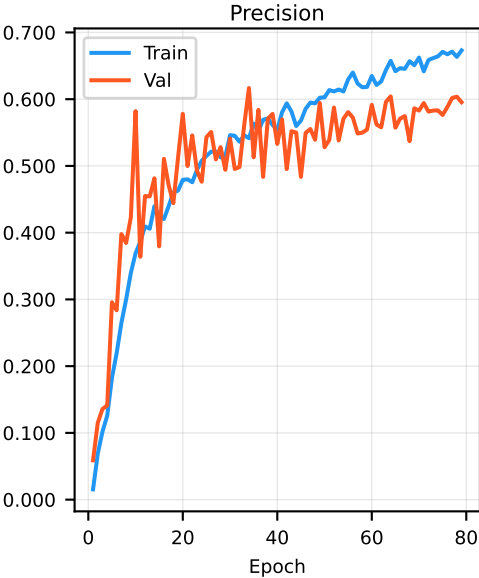
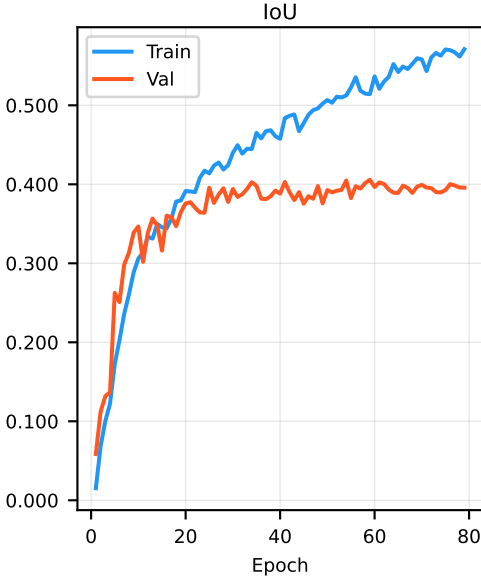
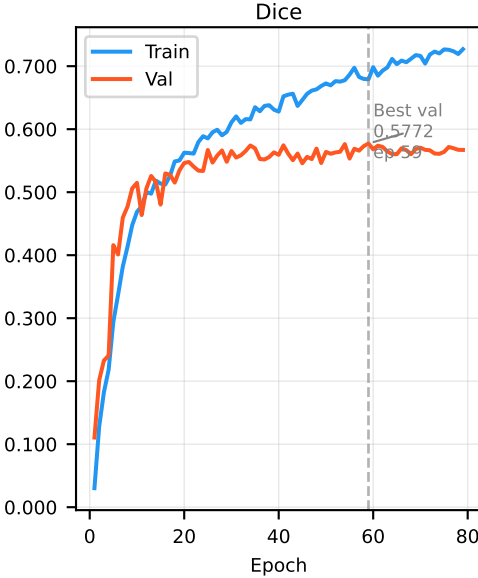
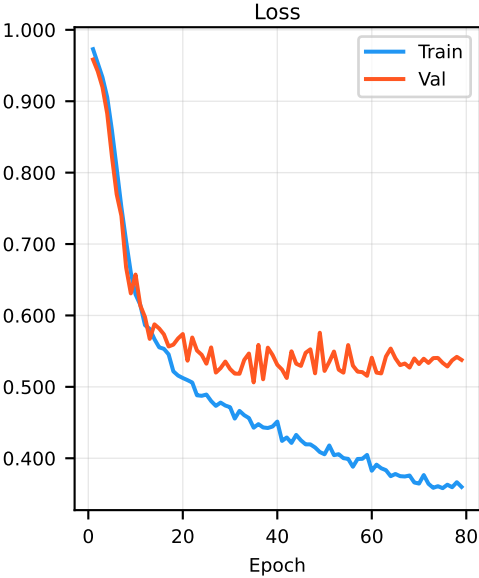
Resumen

Épocas entrenadas: 83
Mejor val Dice : 0.5766
→ Epoch : 63
→ IoU : 0.4051
→ Precision : 0.6357
→ Recall : 0.5275

masses_exp02_resnet34_focalversky
U-Net + RESNET34 (pretrained ImageNet)

Configuración

Parámetro	Valor
Arquitectura	unet
Encoder	resnet34
Encoder weights	imagenet
In channels	1
Image size	1024
Batch size	8
Epochs	100
Aug profile	-
Loss	focal_tversky
alpha	0.7
beta	0.3
gamma	0.75
Optimizer	adamw
LR	5e-05
Weight decay	1e-05
Scheduler	cosine_annealing
T_max	100
Early stop metric	val_dice
Early stop patience	20



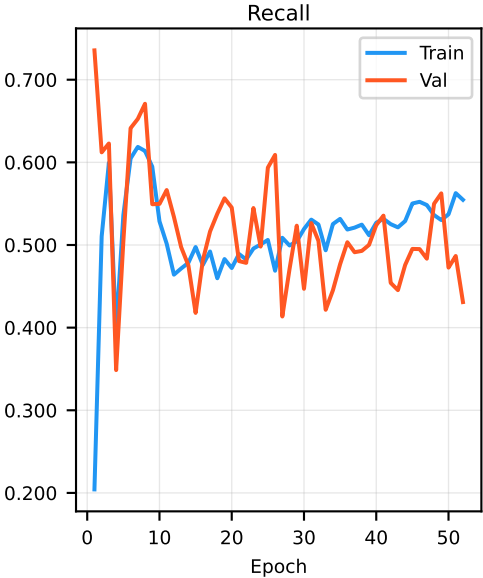
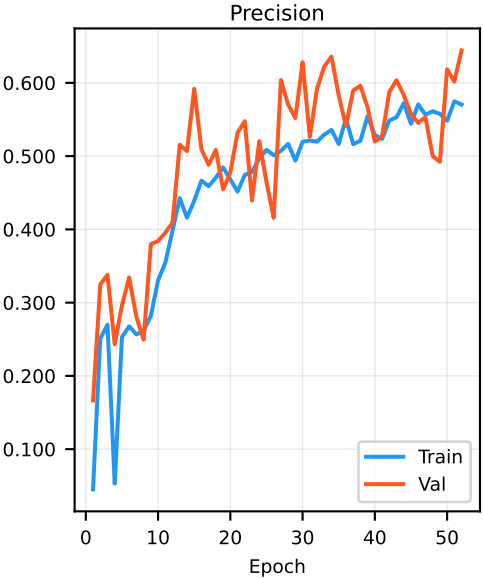
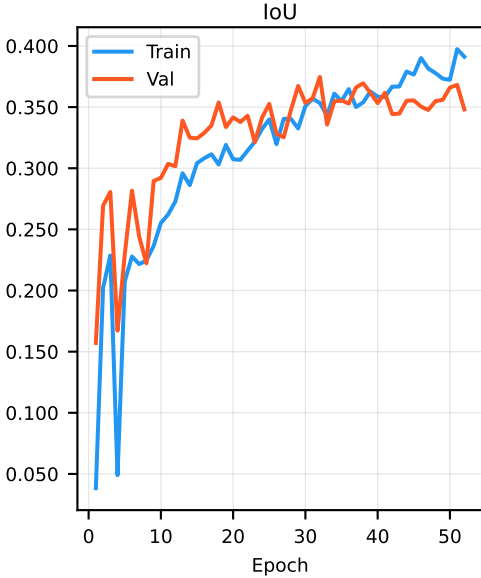
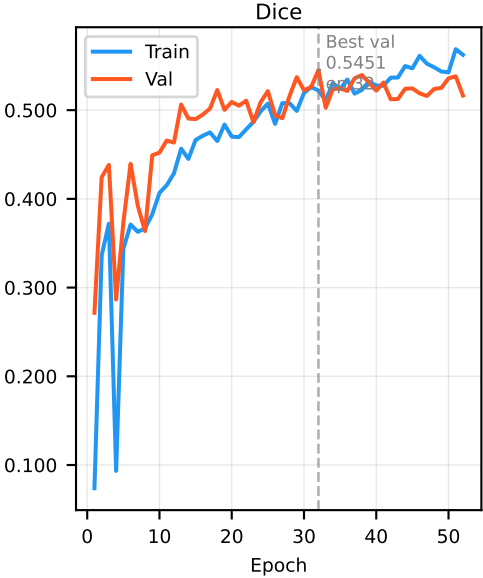
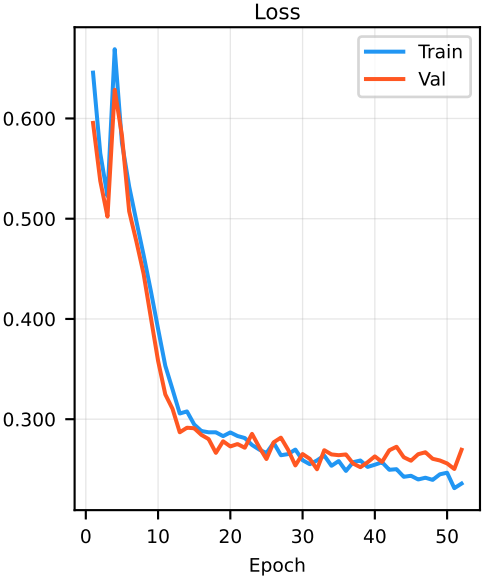
Resumen

Épocas entrenadas: 79
Mejor val Dice : 0.5772
→ Epoch : 59
→ IoU : 0.4057
→ Precision : 0.5544
→ Recall : 0.6019

masses_exp03_resnet50_aggressive_aug
U-Net + RESNET50 (pretrained ImageNet)

Configuración

Parámetro	Valor
Arquitectura	unet
Encoder	resnet50
Encoder weights	imagenet
In channels	1
Image size	1024
Batch size	8
Epochs	100
Aug profile	aggressive
Loss	bce_dice
bce_weight	0.5
dice_weight	0.5
Optimizer	adamw
LR	0.0001
Weight decay	1e-05
Scheduler	cosine_annealing
T_max	100
Early stop metric	val_dice
Early stop pacienc	20



Resumen

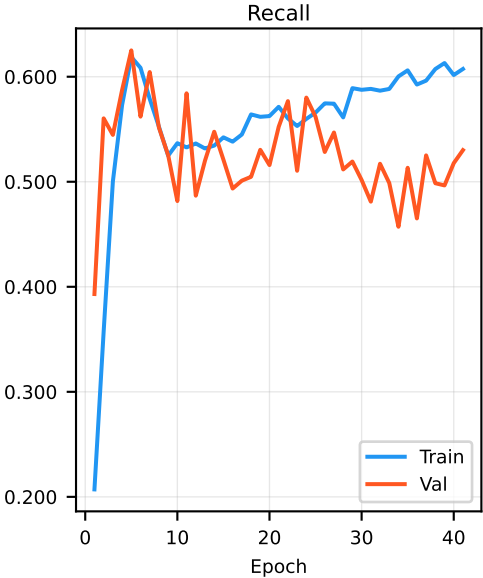
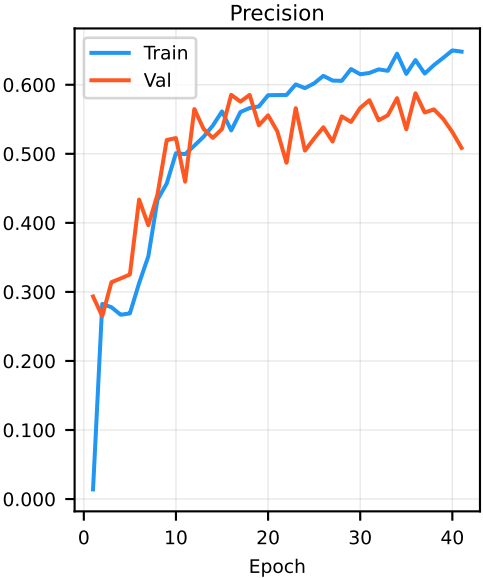
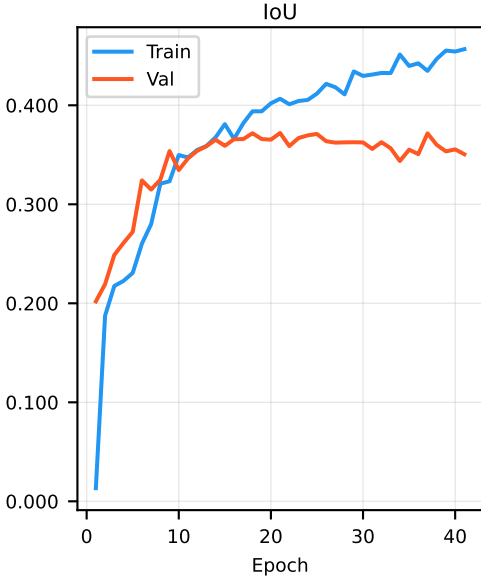
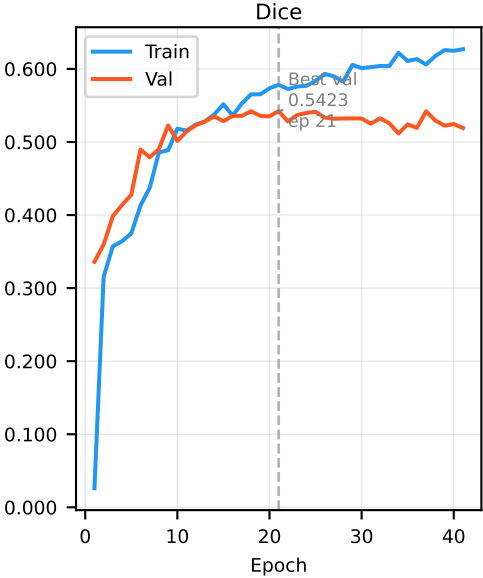
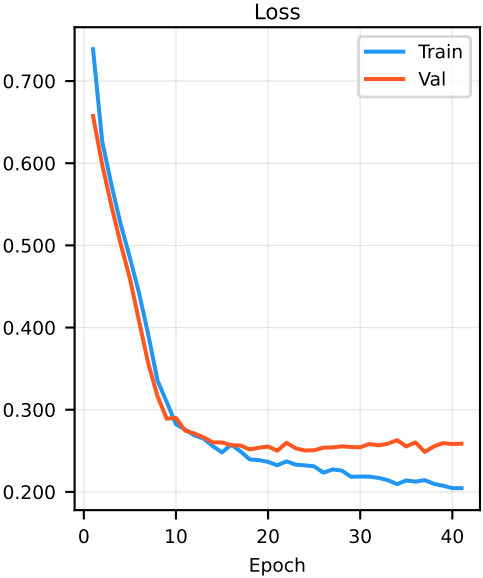
Épocas entrenadas: 52
Mejor val Dice : 0.5451
→ Epoch : 32
→ IoU : 0.3747
→ Precision : 0.5921
→ Recall : 0.5050

masses_exp04_resnet34_finetune

U-Net + RESNET34 (pretrained ImageNet)

Configuración

Parámetro	Valor
Arquitectura	unet
Encoder	resnet34
Encoder weights	imagenet
In channels	1
Image size	1024
Batch size	8
Epochs	100
Aug profile	basic
Loss	bce_dice
bce_weight	0.5
dice_weight	0.5
Optimizer	adamw
LR	-
Weight decay	1e-05
Scheduler	cosine_annealing
T_max	100
Early stop metric	val_dice
Early stop patien	20



Resumen

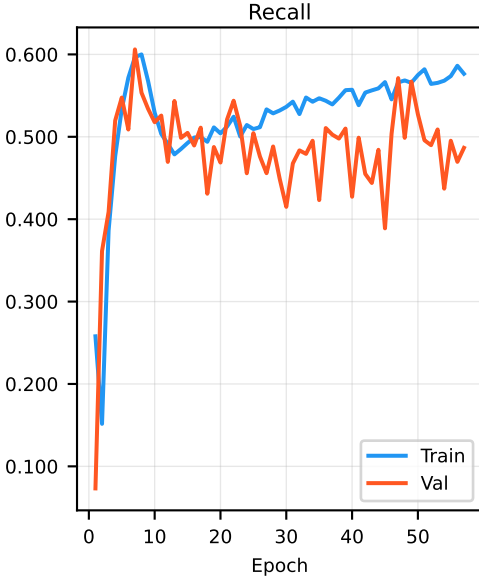
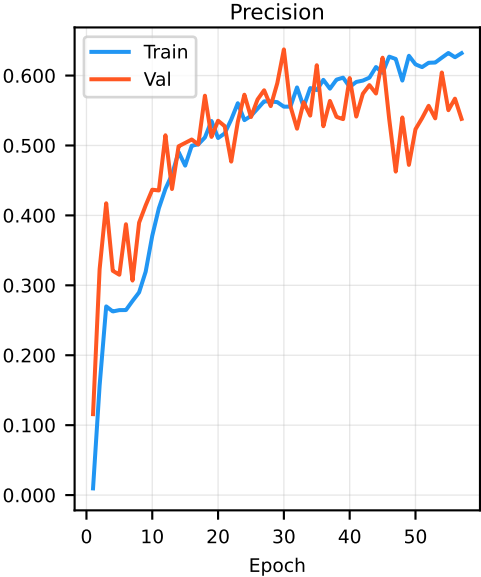
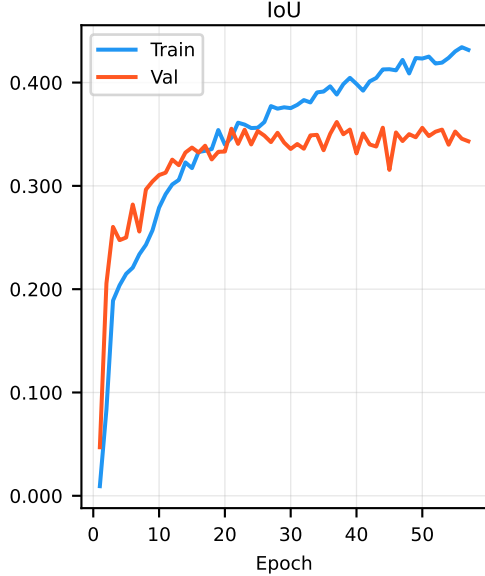
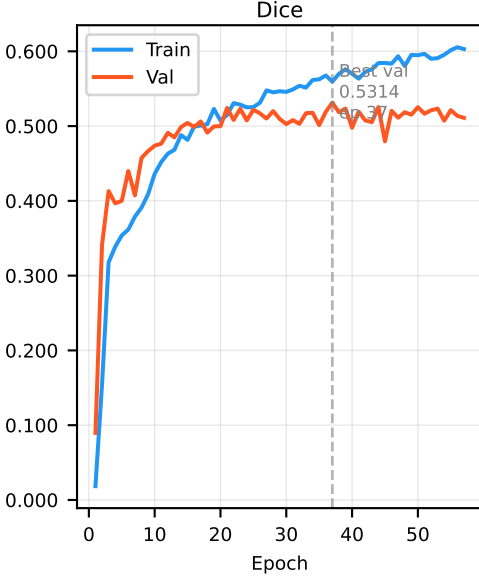
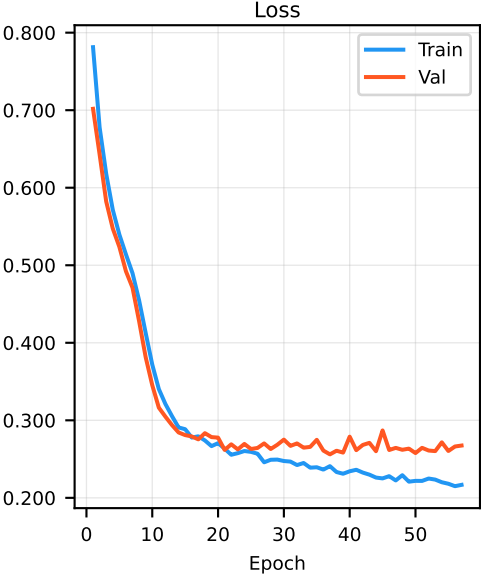
Épocas entrenadas: 41
Mejor val Dice : 0.5423
→ Epoch : 21
→ IoU : 0.3720
→ Precision : 0.5326
→ Recall : 0.5524

masses_exp05_resnet50_freeze3

U-Net + RESNET50 (pretrained ImageNet)

Configuración

Parámetro	Valor
Arquitectura	unet
Encoder	resnet50
Encoder weights	imagenet
In channels	1
Image size	1024
Batch size	8
Epochs	100
Aug profile	basic
Loss	bce_dice
bce_weight	0.5
dice_weight	0.5
Optimizer	adamw
LR	-
Weight decay	1e-05
Scheduler	cosine_annealing
T_max	100
Early stop metric	val_dice
Early stop patienc	20



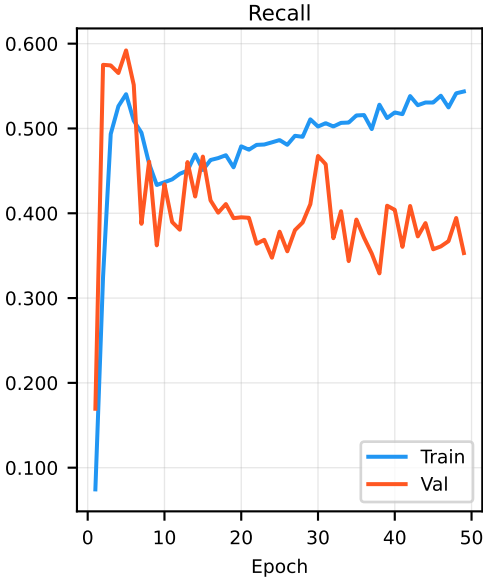
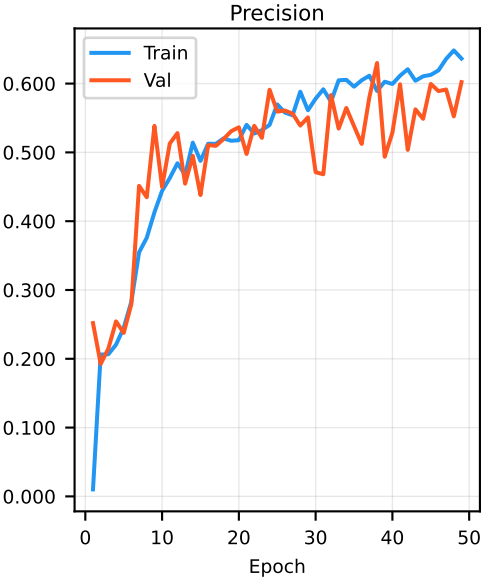
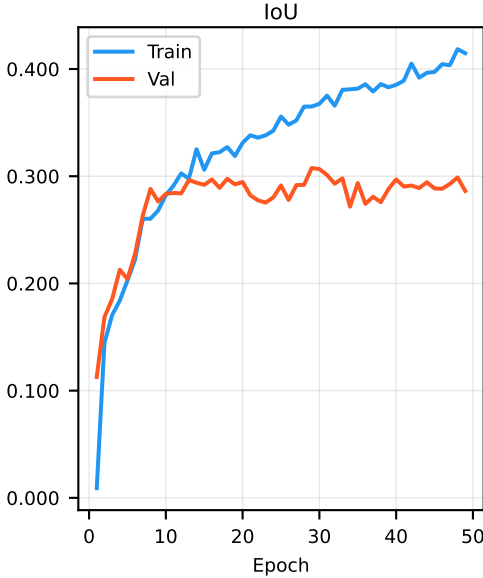
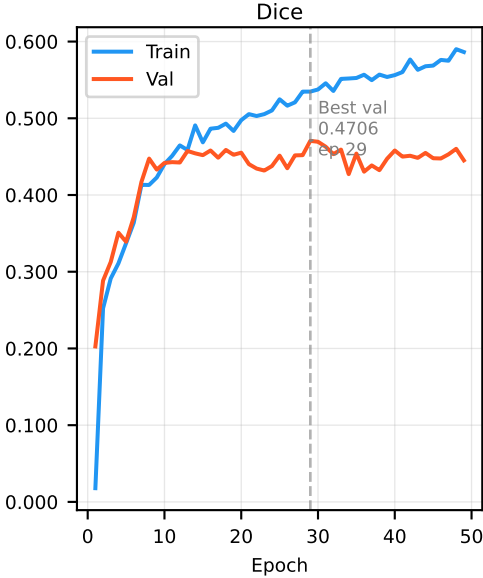
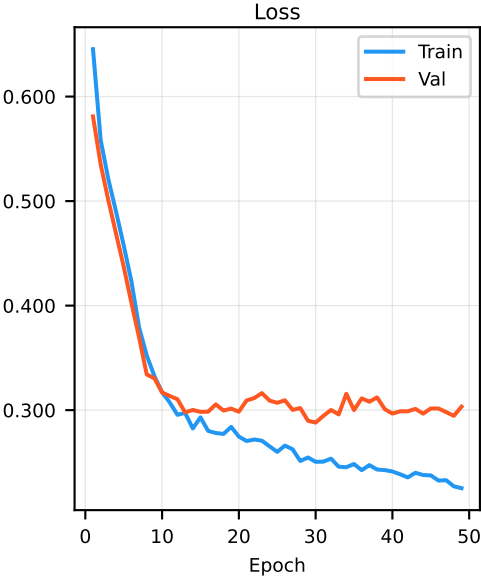
Resumen

Épocas entrenadas: 57
Mejor val Dice : 0.5314
→ Epoch : 37
→ IoU : 0.3619
→ Precision : 0.5638
→ Recall : 0.5026

masses_exp06_resnet34_finetune_clahe
U-Net + RESNET34 (pretrained ImageNet)

Configuración

Parámetro	Valor
Arquitectura	unet
Encoder	resnet34
Encoder weights	imagenet
In channels	1
Image size	1024
Batch size	8
Epochs	100
Aug profile	basic
Loss	bce_dice
bce_weight	0.5
dice_weight	0.5
Optimizer	adamw
LR	-
Weight decay	1e-05
Scheduler	cosine_annealing
T_max	100
Early stop metric	val_dice
Early stop patience	20



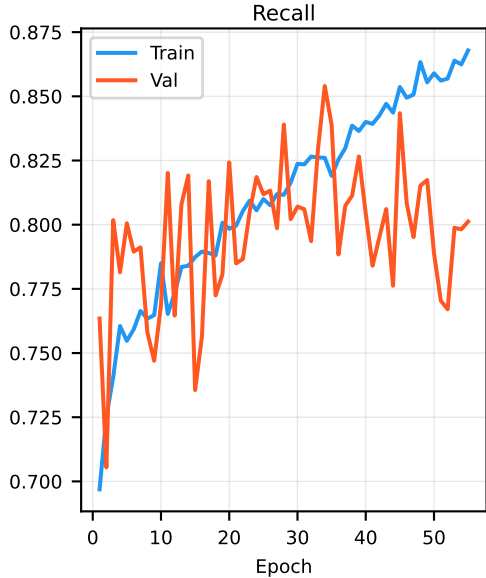
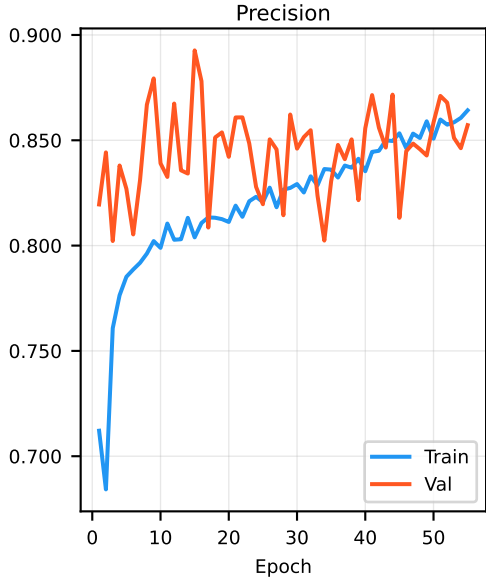
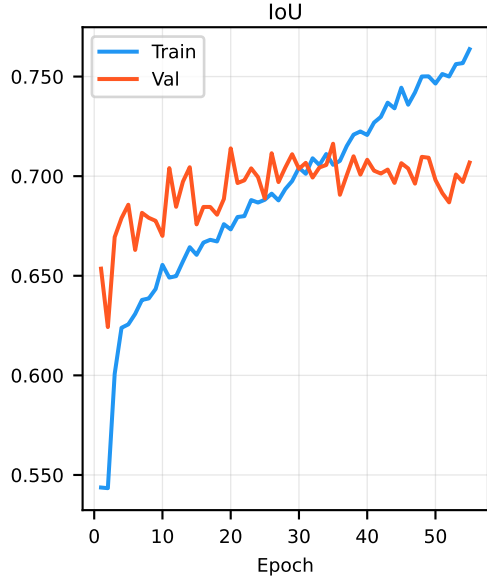
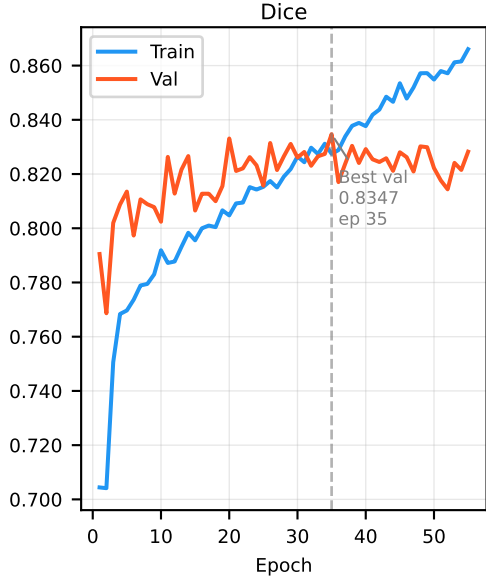
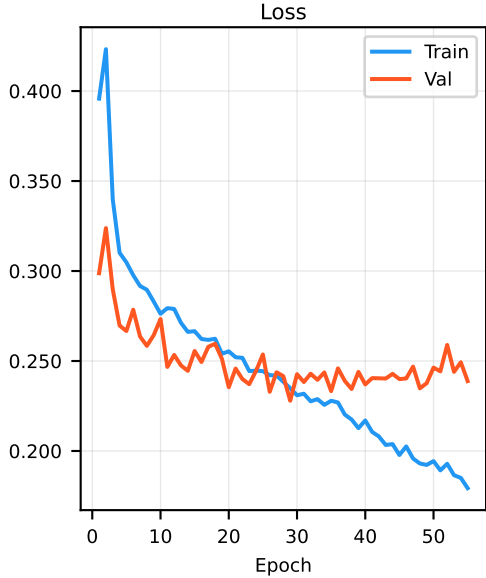
Resumen

Épocas entrenadas: 49
Mejor val Dice : 0.4706
→ Epoch : 29
→ IoU : 0.3077
→ Precision : 0.5509
→ Recall : 0.4107

masses_exp07_patch512_resnet34
U-Net + RESNET34 (pretrained ImageNet)

Configuración

Parámetro	Valor
Arquitectura	unet
Encoder	resnet34
Encoder weights	imagenet
In channels	1
Patch size	512
Centroid jitter	64
Batch size	8
Epochs	100
Aug profile	basic
Loss	bce_dice
bce_weight	0.5
dice_weight	0.5
Optimizer	adamw
LR	0.0001
Weight decay	1e-05
Scheduler	cosine_annealing
T_max	100
Early stop metric	val_dice
Early stop patience	20

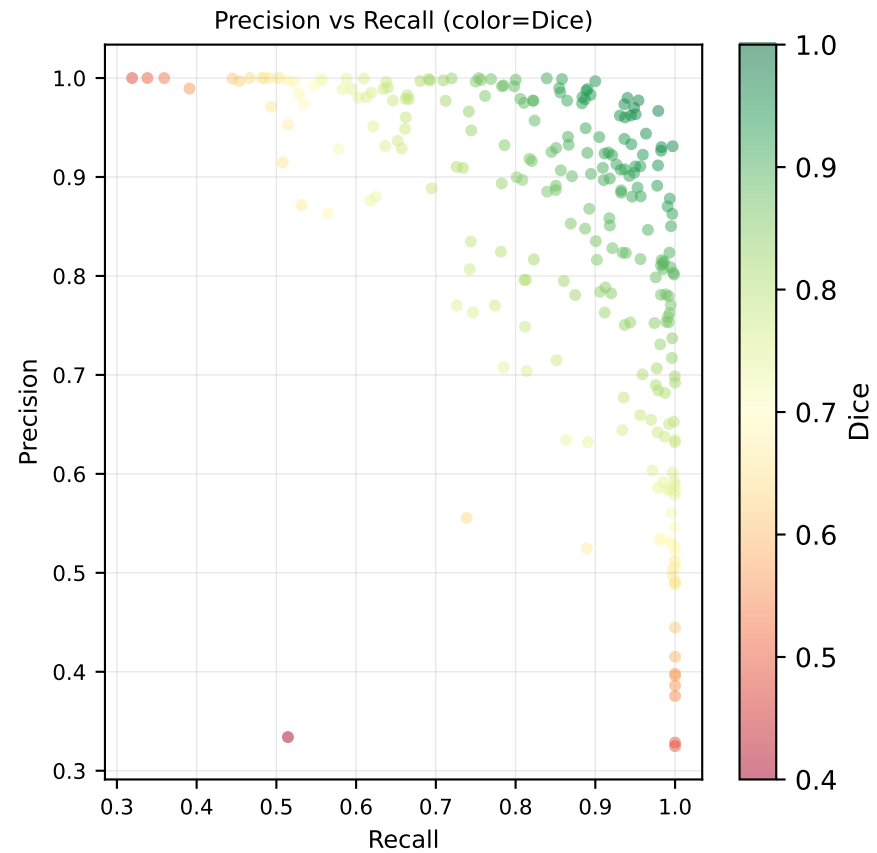
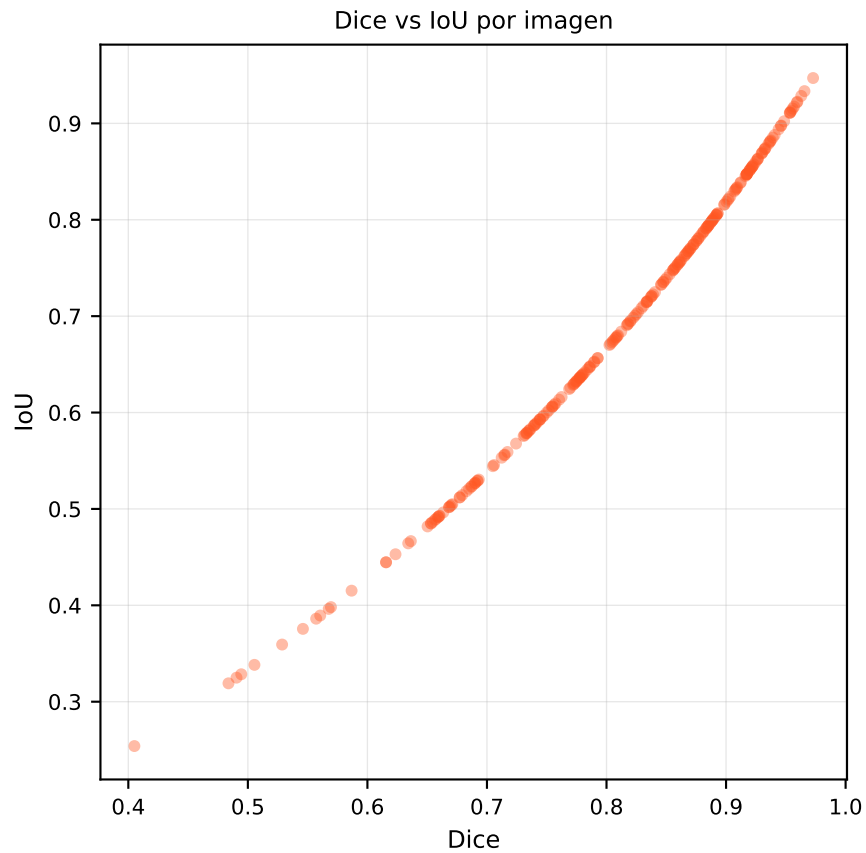
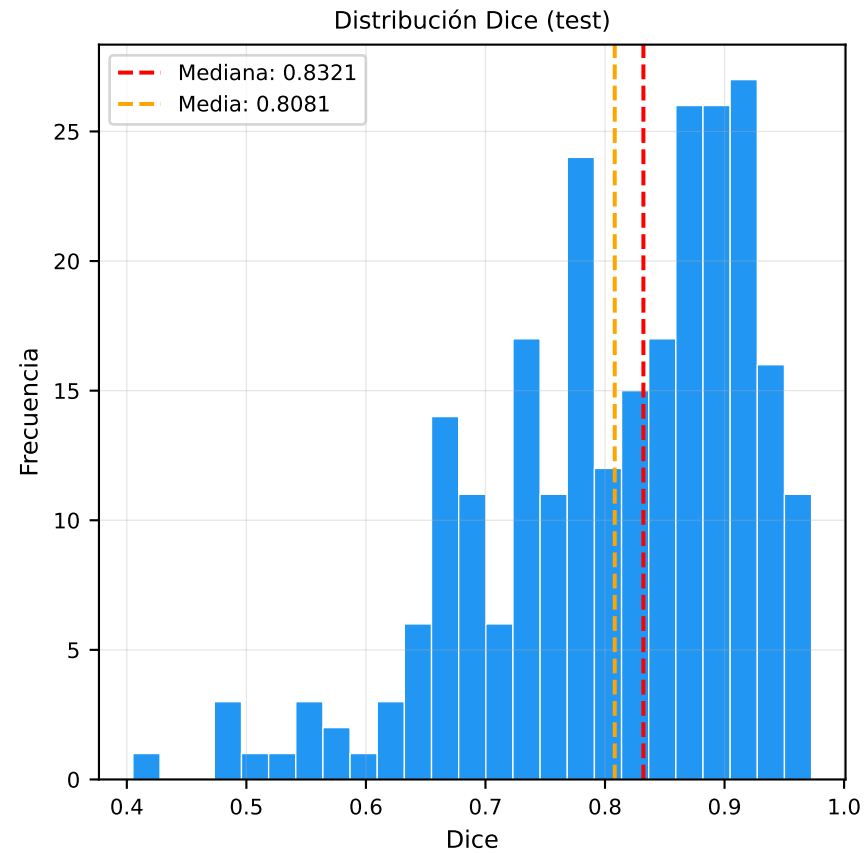


Resumen

Épocas entrenadas: 55
Mejor val Dice : 0.8347
→ Epoch : 35
→ IoU : 0.7163
→ Precision : 0.8306
→ Recall : 0.8389

— TEST —
Val Dice : 0.8347
Imágenes : 254
Duración : 16.3s (0.06s por imagen)
Dice : 0.8093
IoU : 0.6797
Precision : 0.8025
Recall : 0.8162
Dice : 0.8081 ± 0.1097
IoU : 0.6915 ± 0.1469
Precision : 0.8299 ± 0.1688
Recall : 0.8433 ± 0.1651
Dice < 0.5 (fallos graves) : 4 / 254 (1.6%)
Dice > 0.8 : 148 / 254 (58.3%)
Dice > 0.9 (excelentes) : 58 / 254 (22.8%)

masses_exp07_patch512_resnet34 — Detalle Test Set



masses_exp07_patch512_resnet34 — Casos Extremos y Mediana (Test)

Top 5 (mejor Dice)

stem	dice	iou	precision	recall
P_00522_1_MLO	0.9728	0.9471	0.9667	0.979
P_00289_1_MLO	0.9656	0.9335	0.9774	0.9541
P_00462_1_CC_8	0.9629	0.9285	0.9311	0.997
P_00859_1_CC_8	0.9597	0.9225	0.9799	0.9403
P_01226_1_MLO	0.9592	0.9217	0.9701	0.9486

Mediana (~P50)

stem	dice	iou	precision	recall
P_00581_1_MLO	0.8852	0.794	0.9749	0.8106
P_00332_1_CC_8	0.7746	0.6322	0.6375	0.987
P_01047_1_MLO	0.5695	0.3981	0.3981	1.0
P_01485_1_MLO	0.7926	0.6564	0.9825	0.6642
P_00894_1_MLO	0.7771	0.6355	0.7148	0.8513

Bottom 5 (peor Dice)

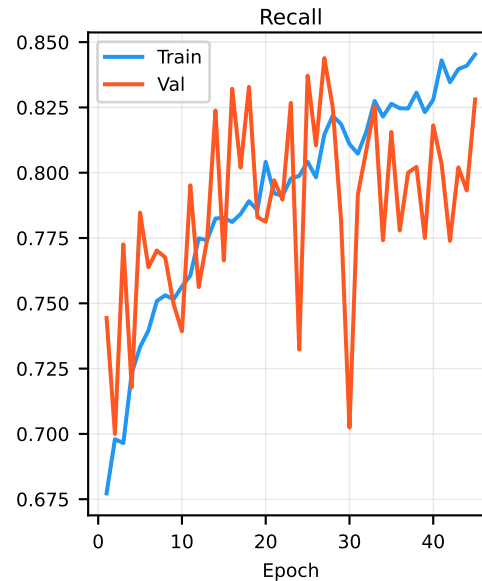
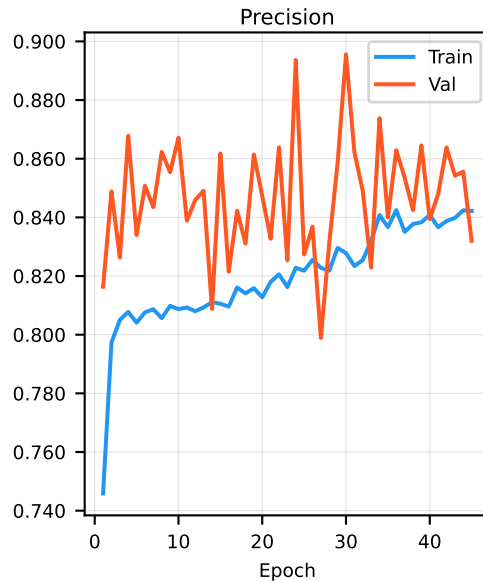
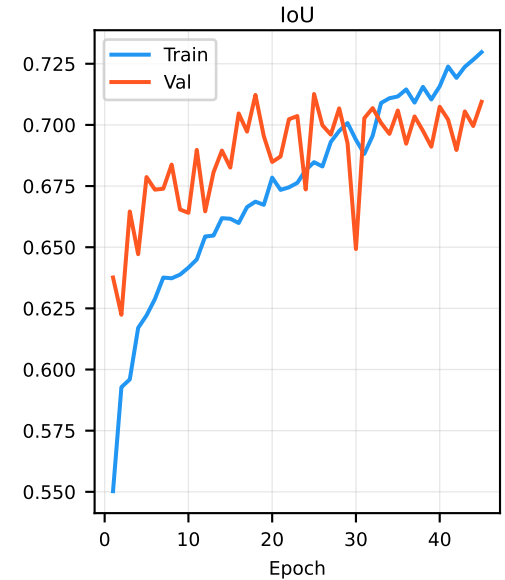
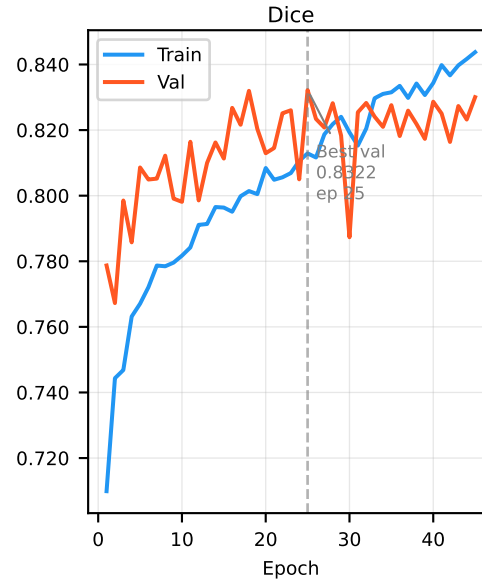
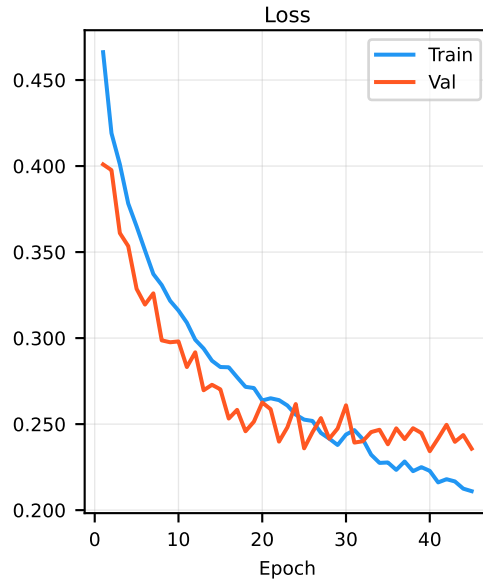
stem	dice	iou	precision	recall
P_00106_2_CC_3	0.4051	0.254	0.334	0.5145
P_00173_1_CC_2	0.4838	0.3191	1.0	0.3191
P_01518_1_CC_7	0.4905	0.3249	0.3249	1.0
P_00207_3_MLO	0.4945	0.3285	0.3285	1.0
P_01768_1_MLO	0.5055	0.3382	1.0	0.3382

masses_exp08_attention_unet_patch512

Attention U-Net + RESNET34 (pretrained ImageNet)

Configuración

Parámetro	Valor
Arquitectura	attention_unet
Encoder	resnet34
Encoder weights	imagenet
In channels	1
Patch size	512
Centroid jitter	64
Batch size	8
Epochs	100
Aug profile	basic
Loss	bce_dice
bce_weight	0.5
dice_weight	0.5
Optimizer	adamw
LR	0.0001
Weight decay	1e-05
Scheduler	cosine_annealing
T_max	100
Early stop metric	val_dice
Early stop patience	20



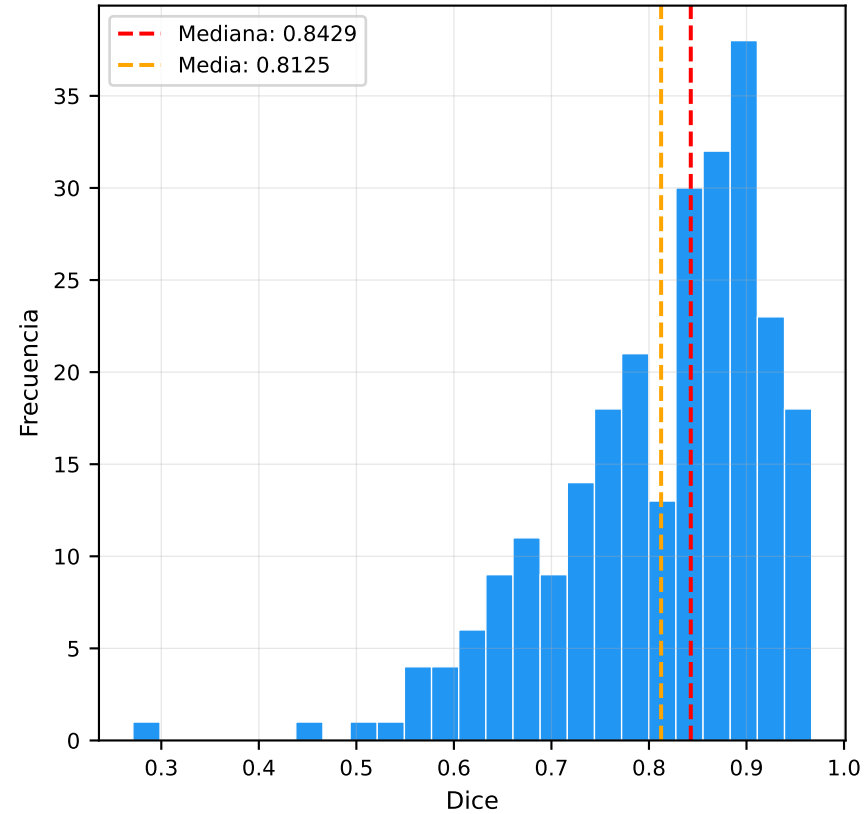
Resumen

Épocas entrenadas: 45
Mejor val Dice : 0.8322
→ Epoch : 25
→ IoU : 0.7126
→ Precision : 0.8273
→ Recall : 0.8372

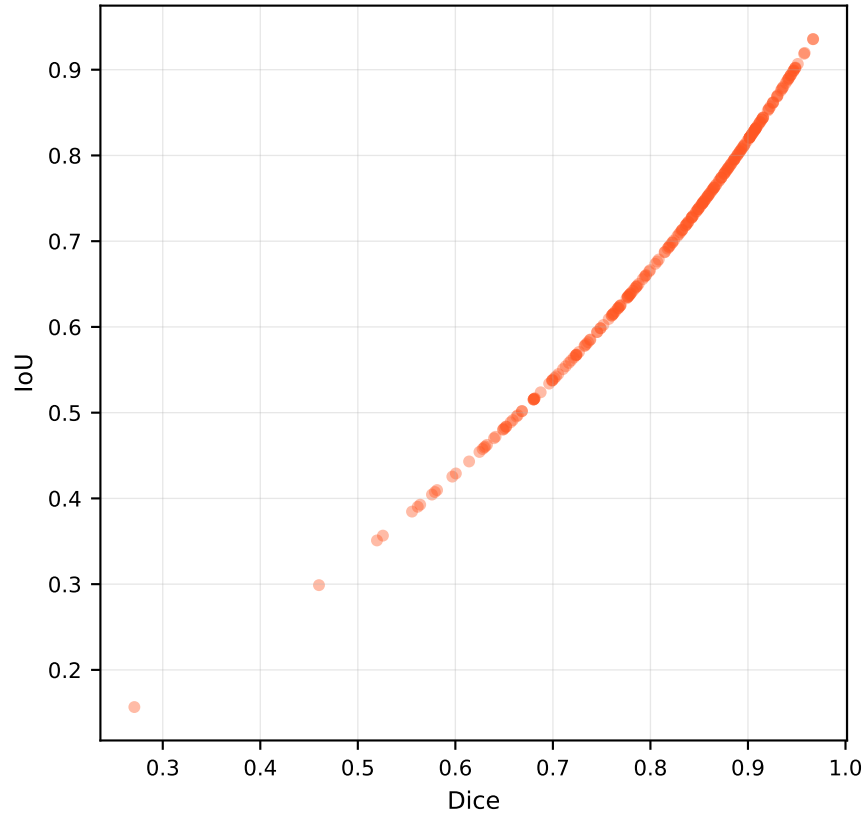
— TEST —
Val Dice : 0.8322
Imágenes : 254
Duración : 14.8s (0.06s por imagen)
Dice : 0.8163
IoU : 0.6896
Precision : 0.8036
Recall : 0.8294
Dice : 0.8125 ± 0.1088
IoU : 0.6972 ± 0.1440
Precision : 0.8065 ± 0.1673
Recall : 0.8696 ± 0.1504
Dice < 0.5 (fallos graves) : 2 / 254 (0.8%)
Dice > 0.8 : 154 / 254 (60.6%)
Dice > 0.9 (excelentes) : 61 / 254 (24.0%)

masses_exp08_attention_unet_patch512 — Detalle Test Set

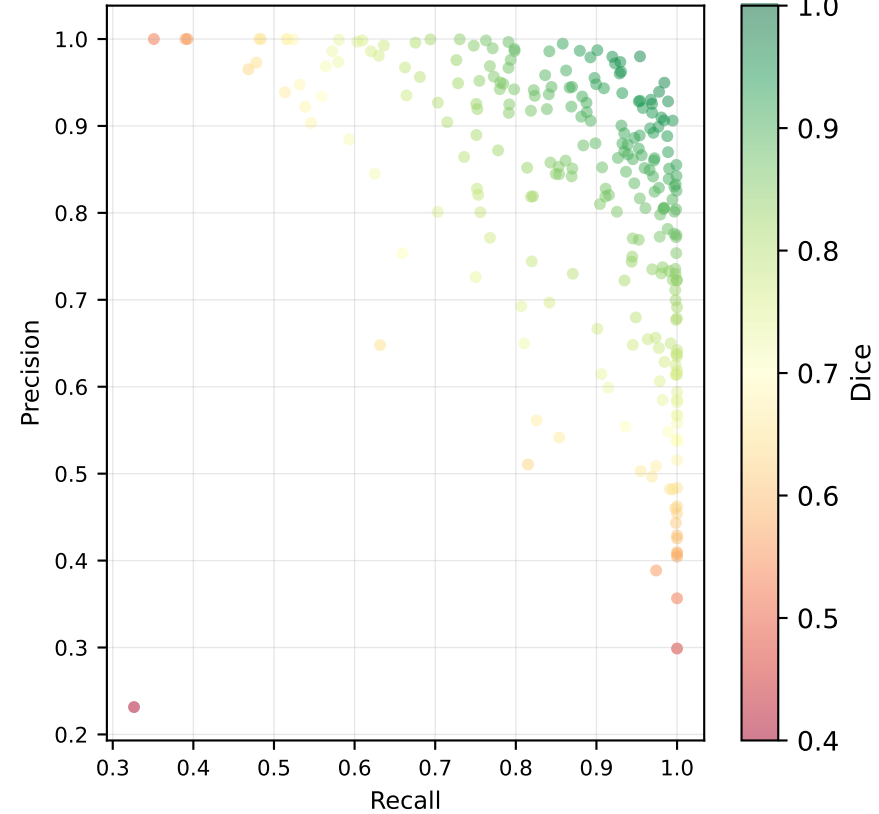
Distribución Dice (test)



Dice vs IoU por imagen



Precision vs Recall (color=Dice)



masses_exp08_attention_unet_patch512 — Casos Extremos y Mediana (Test)

Top 5 (mejor Dice)

stem	dice	iou	precision	recall
P_00703_1_MLO	0.9668	0.9358	0.98	0.954
P_01477_1_CC_2	0.9667	0.9355	0.9498	0.9842
P_01596_1_MLO	0.9581	0.9196	0.9393	0.9777
P_00462_1_CC_8	0.9576	0.9186	0.9282	0.9888
P_01364_1_MLO	0.9511	0.9068	0.9736	0.9296

Mediana (~P50)

stem	dice	iou	precision	recall
P_00581_1_MLO	0.8861	0.7955	0.9364	0.8409
P_00332_1_CC_8	0.6802	0.5154	0.5154	1.0
P_01047_1_MLO	0.6247	0.4542	0.4542	1.0
P_01485_1_MLO	0.8825	0.7897	0.9862	0.7985
P_00894_1_MLO	0.8546	0.7461	0.8101	0.9043

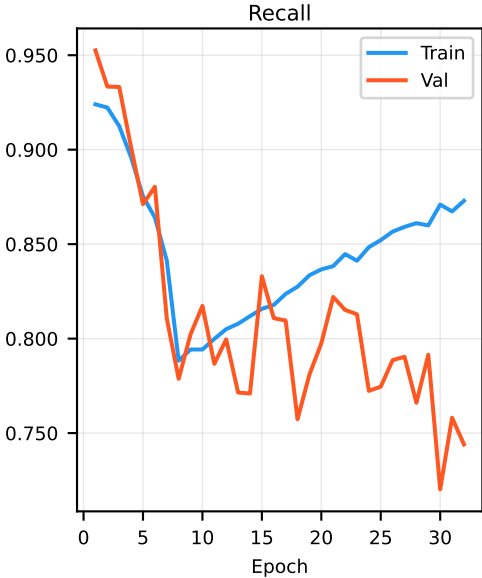
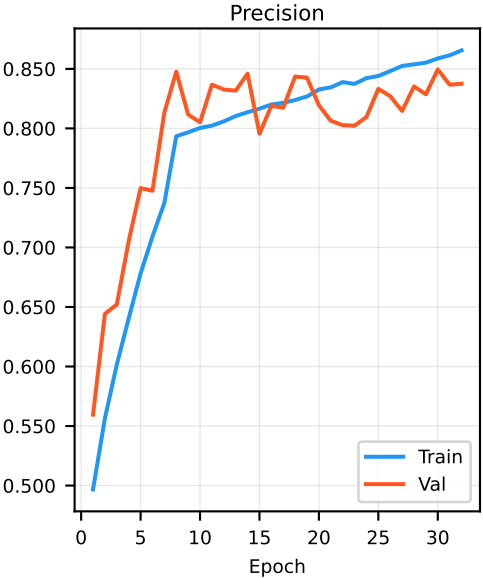
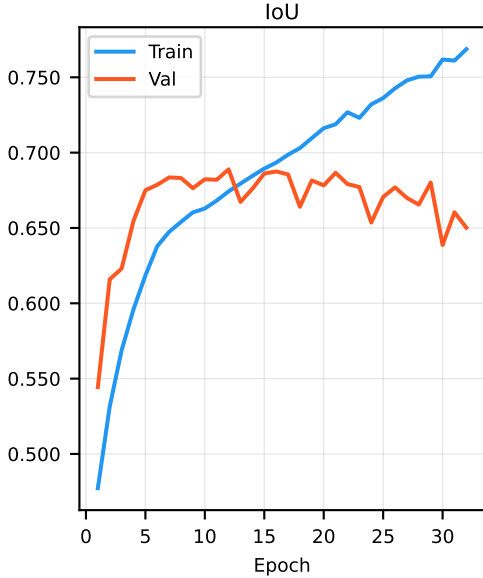
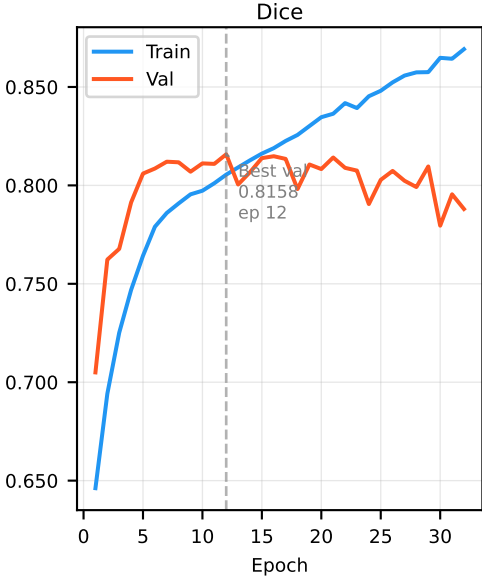
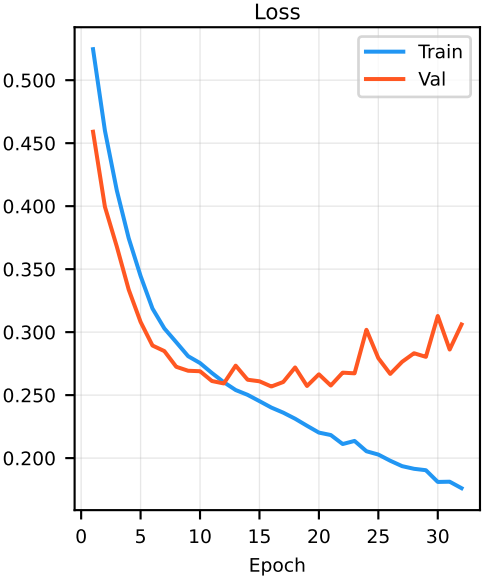
Bottom 5 (peor Dice)

stem	dice	iou	precision	recall
P_00106_2_CC_3	0.2709	0.1567	0.2315	0.3264
P_01518_1_CC_7	0.4602	0.2988	0.2988	1.0
P_00173_1_CC_2	0.5196	0.351	1.0	0.351
P_00218_1_MLO	0.5257	0.3566	0.3566	1.0
P_00224_1_CC_7	0.5556	0.3846	0.3886	0.9741

masses_exp09_attention_unet_3patches
Attention U-Net + RESNET34 (pretrained ImageNet)

Configuración

Parámetro	Valor
Arquitectura	attention_unet
Encoder	resnet34
Encoder weights	imagenet
In channels	1
Patch size	512
Centroid jitter	96
Patches/imagen	3
Batch size	8
Epochs	100
Aug profile	basic
Loss	bce_dice
bce_weight	0.5
dice_weight	0.5
Optimizer	adamw
LR	0.0001
Weight decay	1e-05
Scheduler	cosine_annealing
T_max	100
Early stop metric	val_dice
Early stop pacienc	20



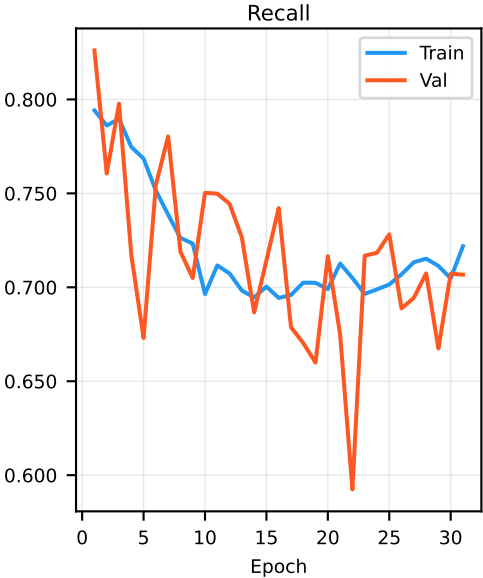
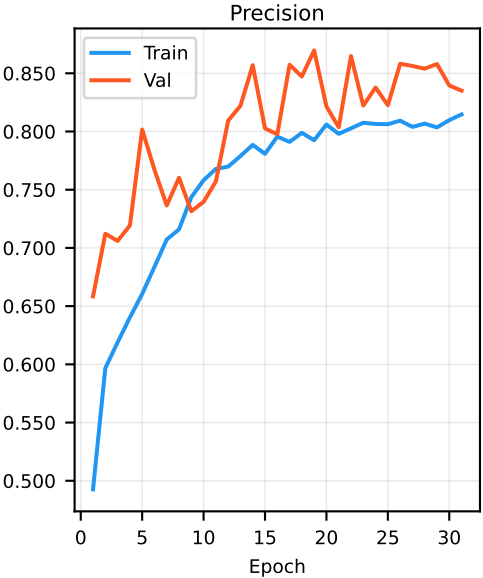
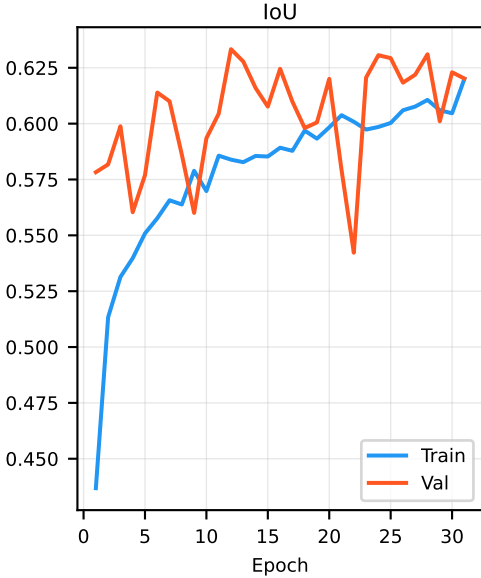
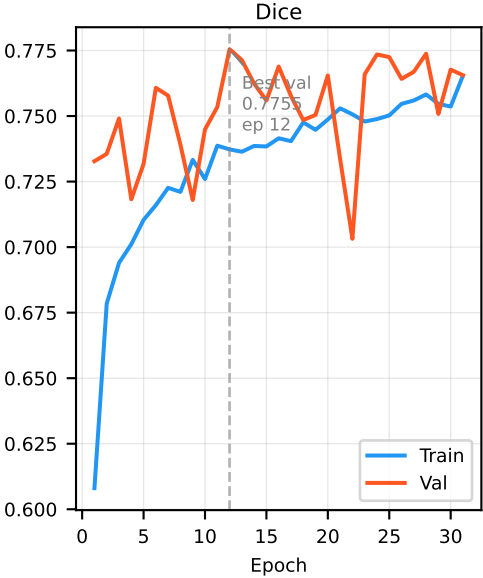
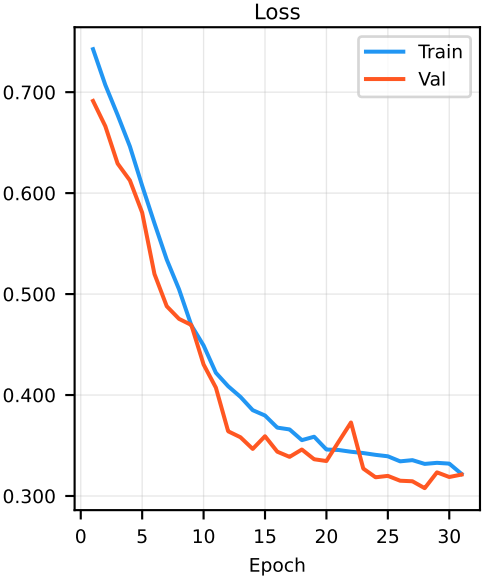
Resumen

Épocas entrenadas: 32
Mejor val Dice : 0.8158
→ Epoch : 12
→ IoU : 0.6889
→ Precision : 0.8326
→ Recall : 0.7996

masses_exp10_attunet_patch768_focaltversky
Attention U-Net + RESNET34 (pretrained ImageNet)

Configuración

Parámetro	Valor
Arquitectura	attention_unet
Encoder	resnet34
Encoder weights	imagenet
In channels	1
Patch size	768
Centroid jitter	96
Patches/imagen	1
Batch size	4
Epochs	100
Aug profile	basic
Loss	focal_tversky
alpha	0.4
beta	0.6
gamma	0.75
Optimizer	adamw
LR	0.0001
Weight decay	1e-05
Scheduler	cosine_annealing
T_max	100
Early stop metric	val_dice
Early stop patience	20



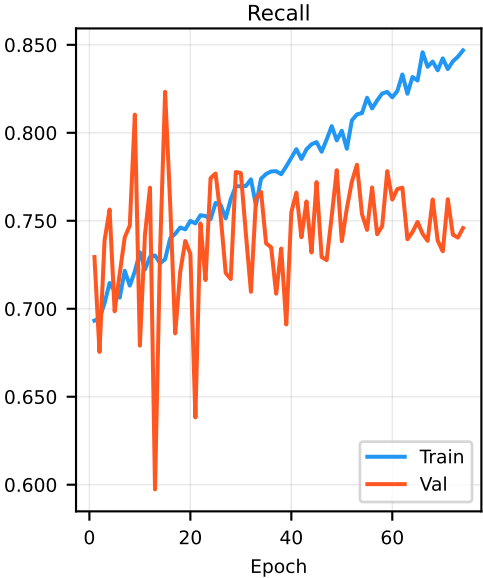
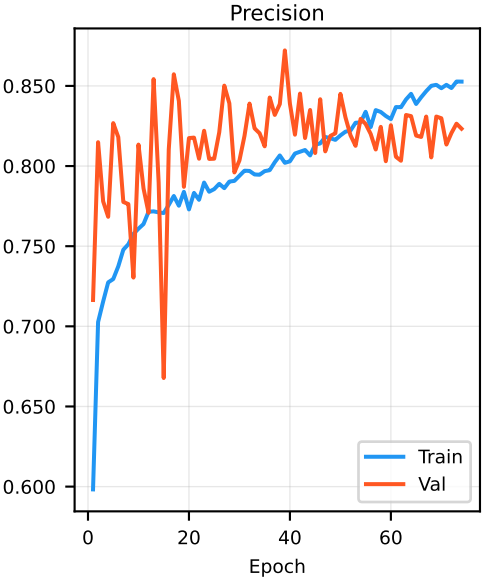
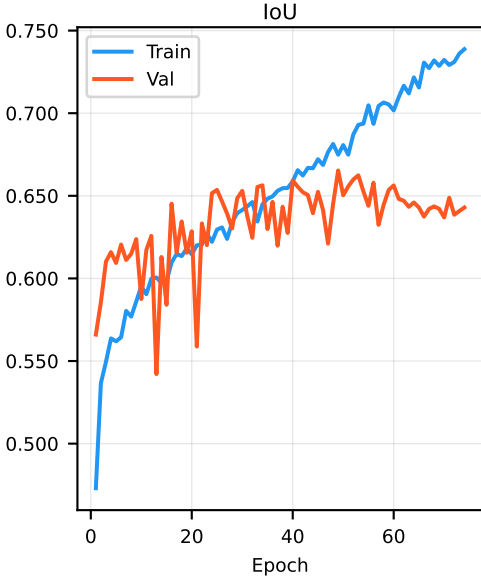
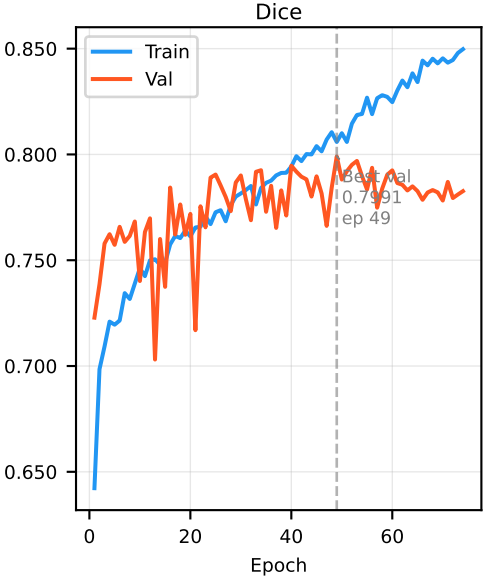
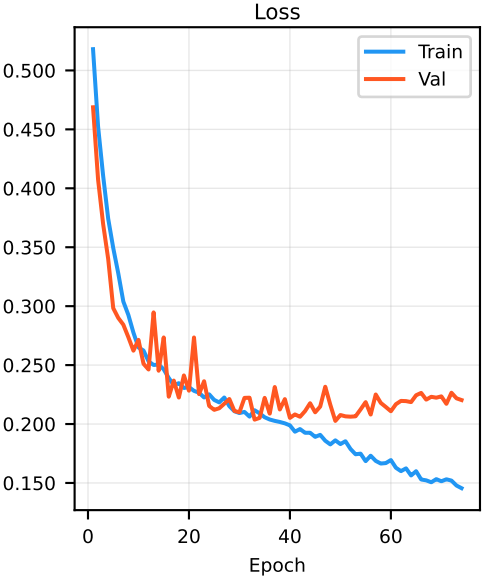
Resumen

Épocas entrenadas: 31
Mejor val Dice : 0.7755
→ Epoch : 12
→ IoU : 0.6333
→ Precision : 0.8093
→ Recall : 0.7445

masses_exp11_attunet_patch768
Attention U-Net + RESNET34 (pretrained ImageNet)

Configuración

Parámetro	Valor
Arquitectura	attention_unet
Encoder	resnet34
Encoder weights	imagenet
In channels	1
Patch size	768
Centroid jitter	96
Patches/imagen	1
Batch size	4
Epochs	100
Aug profile	basic
Loss	bce_dice
bce_weight	0.5
dice_weight	0.5
Optimizer	adamw
LR	0.0001
Weight decay	1e-05
Scheduler	cosine_annealing
T_max	100
Early stop metric	val_dice
Early stop pacienc	25



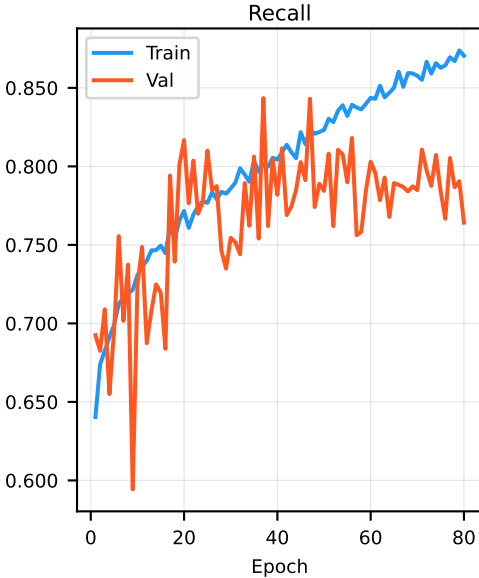
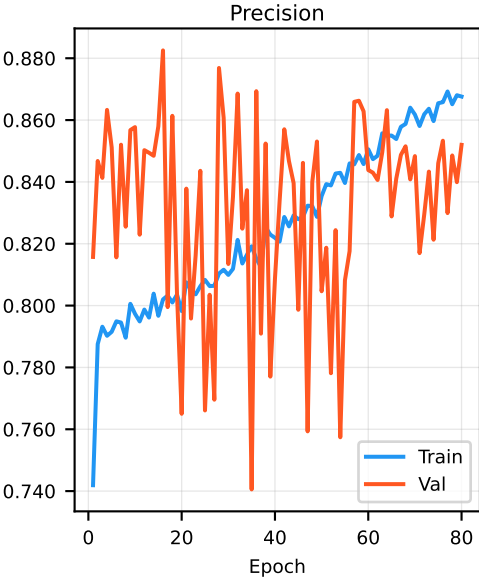
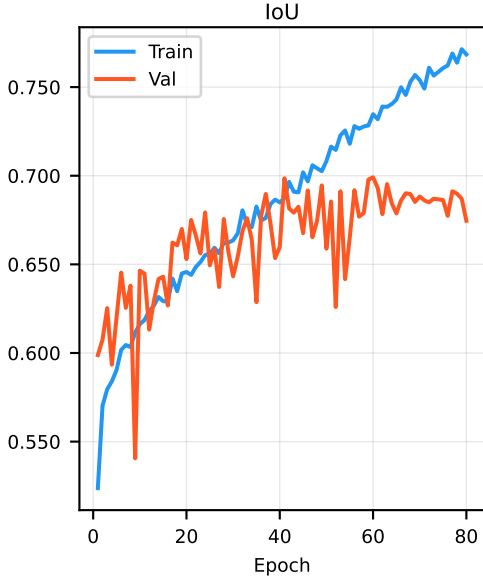
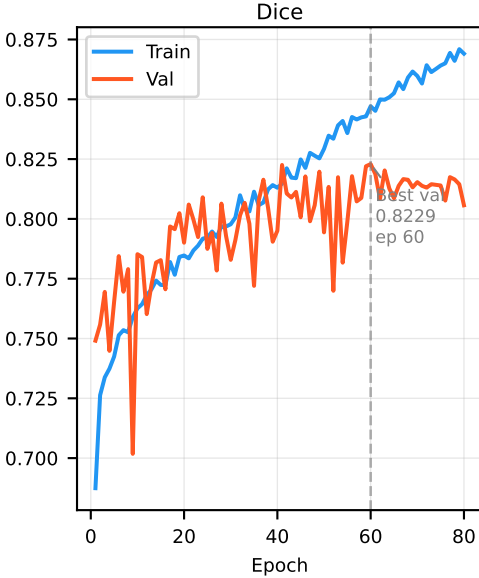
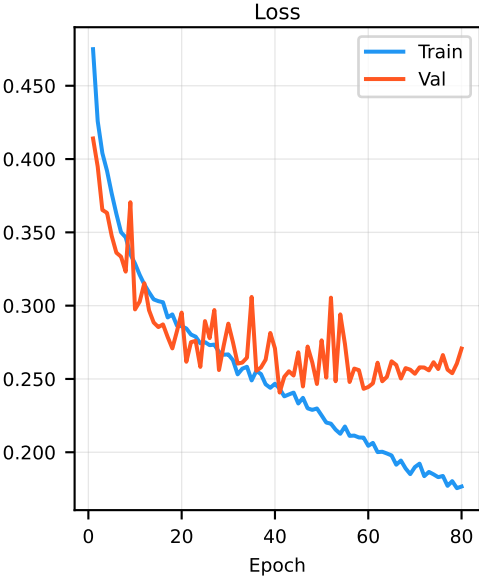
Resumen

Épocas entrenadas: 74
Mejor val Dice : 0.7991
→ Epoch : 49
→ IoU : 0.6654
→ Precision : 0.8204
→ Recall : 0.7788

masses_exp12_attunet_patch512_clahe
Attention U-Net + RESNET34 (pretrained ImageNet)

Configuración

Parámetro	Valor
Arquitectura	attention_unet
Encoder	resnet34
Encoder weights	imagenet
In channels	1
Patch size	512
Centroid jitter	64
Patches/imagen	1
Batch size	8
Epochs	100
Aug profile	basic
Loss	bce_dice
bce_weight	0.5
dice_weight	0.5
Optimizer	adamw
LR	0.0001
Weight decay	1e-05
Scheduler	cosine_annealing
T_max	100
Early stop metric	val_dice
Early stop pacienc	20



Resumen

Épocas entrenadas: 80
Mejor val Dice : 0.8229
→ Epoch : 60
→ IoU : 0.6991
→ Precision : 0.8438
→ Recall : 0.8030

2. Clasificación de Malignidad

EfficientNet-B0 / B2 / B3 • 2 canales: imagen + mapa de probabilidad segmentador

Resumen — Experimentos de Clasificación

Experimento	Backbone	Dropout	Freeze blocks	Epochs	LR backbone	LR classifier
classifier_exp01_efficientnet_l	efficientnet_b3	0.3	0	50	1e-05	0.0001
classifier_exp02_efficientnet_l	efficientnet_b3	0.5	4	50	5e-06	0.0001
classifier_exp03_efficientnet_l	efficientnet_b0	0.4	2	50	1e-05	0.0001
classifier_exp04_b0_weighted	efficientnet_b0	0.4	2	60	1e-05	0.0001
classifier_exp05_b2_aggressive	efficientnet_b2	0.5	2	100	5e-06	0.0003

NOTA: Los experimentos de clasificación no guardan metrics.csv.
Solo se dispone de checkpoints. Las métricas de entrenamiento/validación
pueden recuperarse ejecutando el clasificador con --eval sobre el checkpoint guardado.

classifier_exp01_efficientnet_b3
EfficientNet-B3 ~11M params

Configuración

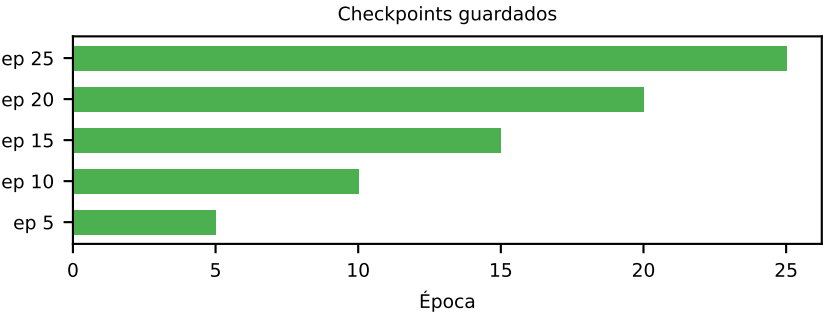
Parámetro	Valor
Backbone	efficientnet_b3
In channels	2
Dropout	0.3
Freeze blocks	0
Pretrained	True
Patch size	512
Batch size	16
Epochs	50
Disc. LR	True
LR backbone	1e-05
LR classifier	0.0001
Weight decay	1e-05
Scheduler	cosine_annealing
ES metric	val_auc
ES patience	15
Segmentador checkpoint	best.pt

Segmentador asociado + Flujo de datos

Flujo de inferencia (clasificación):

1. Imagen mamografía (PNG uint16)
2. Extraer patch 512×512 centrado en lesión
3. Segmentador (Attention U-Net)
→ mapa de probabilidad [0,1]
4. Stack: [imagen | prob_map] → 2 canales
5. EFFICIENTNET_B3 → 2 clases
(BENIGN / MALIGNANT)

Segmentador: best.pt



classifier_exp02_efficientnet_b3_regularized

EfficientNet-B3 ~11M params

Configuración

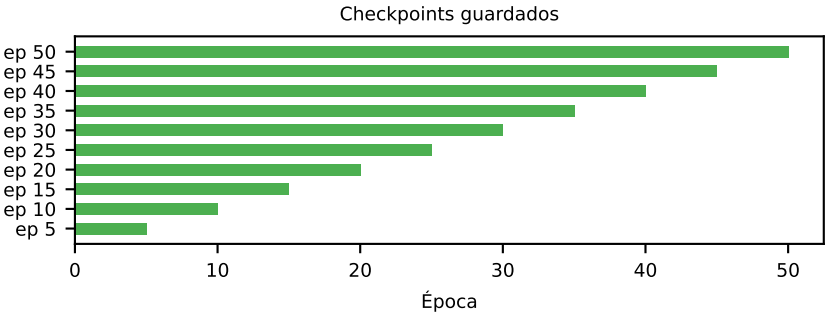
Parámetro	Valor
Backbone	efficientnet_b3
In channels	2
Dropout	0.5
Freeze blocks	4
Pretrained	True
Patch size	512
Batch size	16
Epochs	50
Disc. LR	True
LR backbone	5e-06
LR classifier	0.0001
Weight decay	0.0001
Scheduler	cosine_annealing
ES metric	val_auc
ES patience	15
Segmentador checkpoint	best.pt

Segmentador asociado + Flujo de datos

Flujo de inferencia (clasificación):

1. Imagen mamografía (PNG uint16)
2. Extraer patch 512×512 centrado en lesión
3. Segmentador (Attention U-Net)
→ mapa de probabilidad [0,1]
4. Stack: [imagen | prob_map] → 2 canales
5. EFFICIENTNET_B3 → 2 clases
(BENIGN / MALIGNANT)

Segmentador: best.pt



classifier_exp03_efficientnet_b0
EfficientNet-B0 (Tan & Le, 2019) ~4M params

Configuración

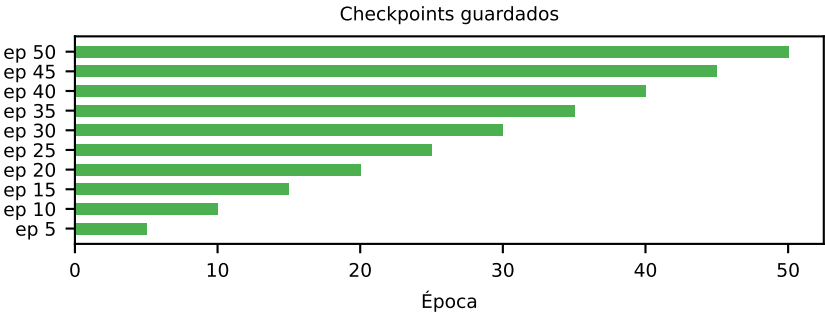
Parámetro	Valor
Backbone	efficientnet_b0
In channels	2
Dropout	0.4
Freeze blocks	2
Pretrained	True
Patch size	512
Batch size	16
Epochs	50
Disc. LR	True
LR backbone	1e-05
LR classifier	0.0001
Weight decay	5e-05
Scheduler	cosine_annealing
ES metric	val_auc
ES patience	15
Segmentador checkpoint	best.pt

Segmentador asociado + Flujo de datos

Flujo de inferencia (clasificación):

1. Imagen mamografía (PNG uint16)
2. Extraer patch 512×512 centrado en lesión
3. Segmentador (Attention U-Net)
→ mapa de probabilidad [0,1]
4. Stack: [imagen | prob_map] → 2 canales
5. EFFICIENTNET_B0 → 2 clases
(BENIGN / MALIGNANT)

Segmentador: best.pt



classifier_exp04_b0_weighted_f1
EfficientNet-B0 (Tan & Le, 2019) ~4M params

Configuración

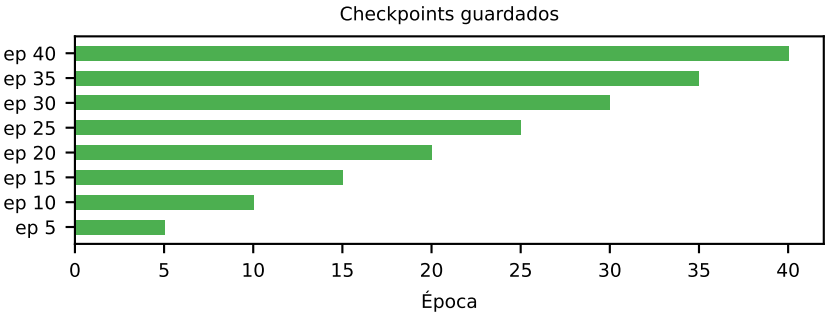
Parámetro	Valor
Backbone	efficientnet_b0
In channels	2
Dropout	0.4
Freeze blocks	2
Pretrained	True
Patch size	512
Batch size	16
Epochs	60
Disc. LR	True
LR backbone	1e-05
LR classifier	0.0001
Weight decay	5e-05
Scheduler	cosine_annealing
ES metric	val_f1
ES patience	15
Segmentador checkpoint	best.pt

Segmentador asociado + Flujo de datos

Flujo de inferencia (clasificación):

1. Imagen mamografía (PNG uint16)
2. Extraer patch 512×512 centrado en lesión
3. Segmentador (Attention U-Net)
→ mapa de probabilidad [0,1]
4. Stack: [imagen | prob_map] → 2 canales
5. EFFICIENTNET_B0 → 2 clases
(BENIGN / MALIGNANT)

Segmentador: best.pt



classifier_exp05_b2_aggressive_aug
EfficientNet-B2 ~9M params

Configuración

Parámetro	Valor
Backbone	efficientnet_b2
In channels	2
Dropout	0.5
Freeze blocks	2
Pretrained	True
Patch size	512
Batch size	16
Epochs	100
Disc. LR	True
LR backbone	5e-06
LR classifier	0.0003
Weight decay	0.0001
Scheduler	cosine_annealing
ES metric	val_f1
ES patience	25
Segmentador checkpoint	best.pt

Segmentador asociado + Flujo de datos

Flujo de inferencia (clasificación):

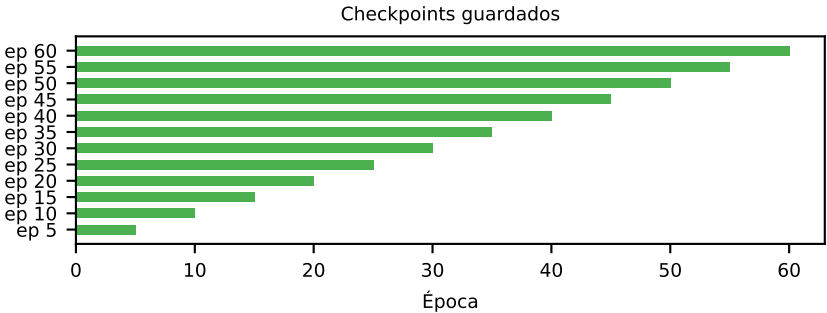
- Imagen mamografía (PNG uint16)
- Extraer patch 512×512 centrado en lesión
- Segmentador (Attention U-Net)
→ mapa de probabilidad [0,1]
- Stack: [imagen | prob_map] → 2 canales
- EFFICIENTNET_B2 → 2 clases
(BENIGN / MALIGNANT)

Segmentador: best.pt

Augmentaciones:

```

horizontal_flip: 0.5
vertical_flip: 0.3
random_rotate90: 0.5
shift_scale_rotate: {'shift_limit': 0.1, 'scale_limit': 0.15, 'rotate_limit': 15, 'p': 0.5}
elastic_transform: {'alpha': 1.0, 'sigma': 50.0, 'p': 0.3}
grid_distortion: {'num_steps': 5, 'distort_limit': 0.3, 'p': 0.2}
brightness_contrast: {'brightness_limit': 0.15, 'contrast_limit': 0.15, 'p': 0.4}
random_gamma: {'gamma_limit': [80, 120], 'p': 0.3}
  
```



3. Fusión Multimodal: Imagen + Radiómicas

EfficientNet-B2 + PyRadiomics • MLP de fusión • Dataset: CBIS-DDSM masses

fusion_exp01_b2_radiomics
EfficientNet-B2 + PyRadiomics (Fusión Multimodal)

Configuración

Parámetro	Valor
Backbone	efficientnet_b2
In channels	2
n_radiomics	107
Rad threshold	0.5
Dropout 1	0.4
Dropout 2	0.3
Patch size	512
Batch size	16
Epochs	80
LR backbone	1e-05
LR fusion	0.0001
Weight decay	0.0001
Scheduler	cosine_annealing
Class weights	[1.0, 1.5]
Label smoothing	0.1
ES metric	val_f1
ES patience	15
Segmentador checkpoint	best.pt
Backbone checkpoint	best.pt

Métricas — Test Set

Checkpoint : /workspace/cbis-unet-mammo/runs/fusion_exp01_b2_radiomics/checkpoints/best.pt
Val AUC : 0.8500
Val F1 : 0.7968
Segmentador : val_dice=0.8322
Imágenes test : 254 (MALIGNANT=122 | BENIGN=132)

AUC-ROC

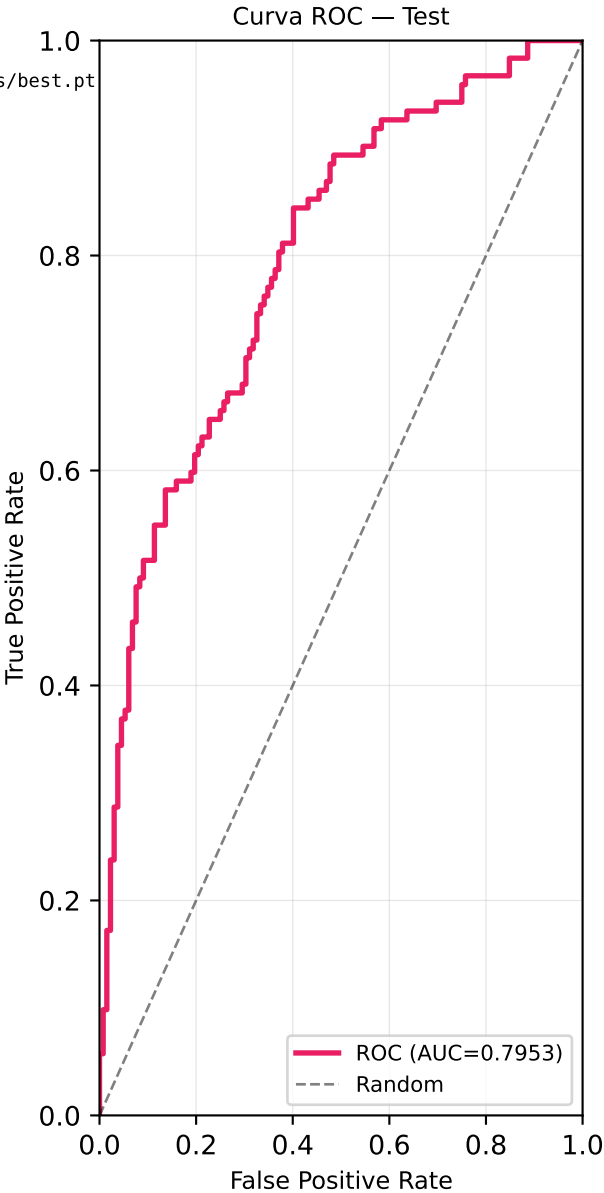
AUC-ROC : 0.7953

Accuracy : 0.6969
F1 : 0.6908
Precision : 0.6772
Recall : 0.7049

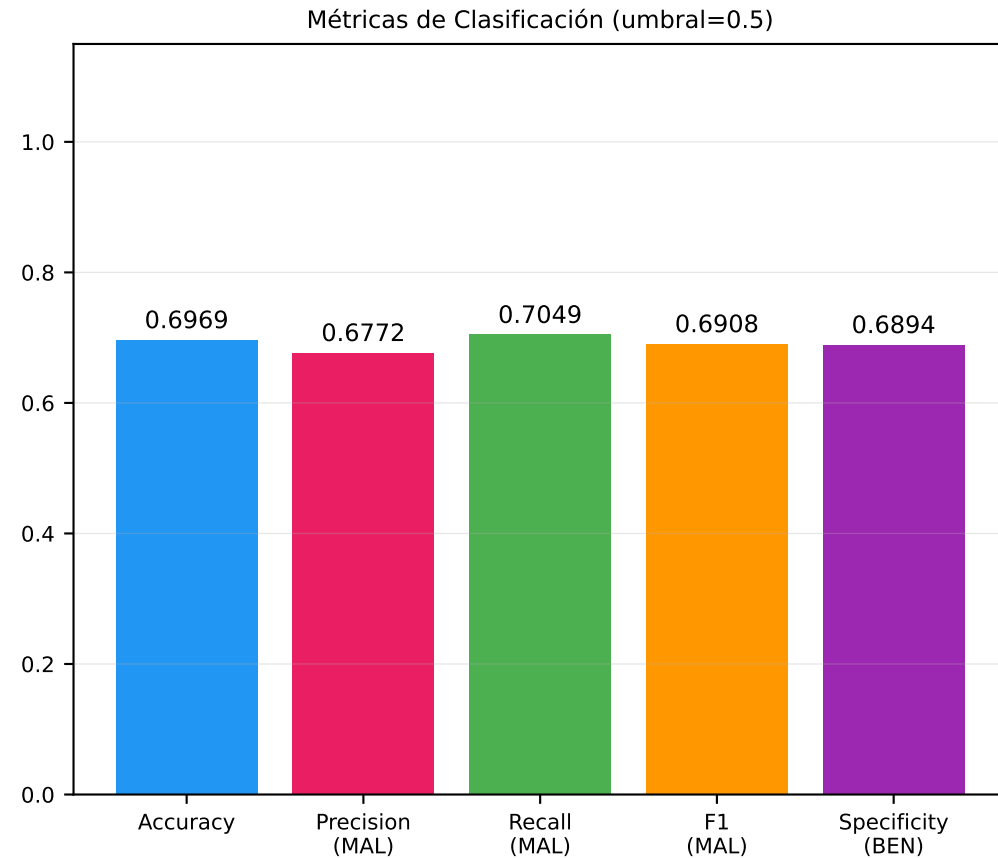
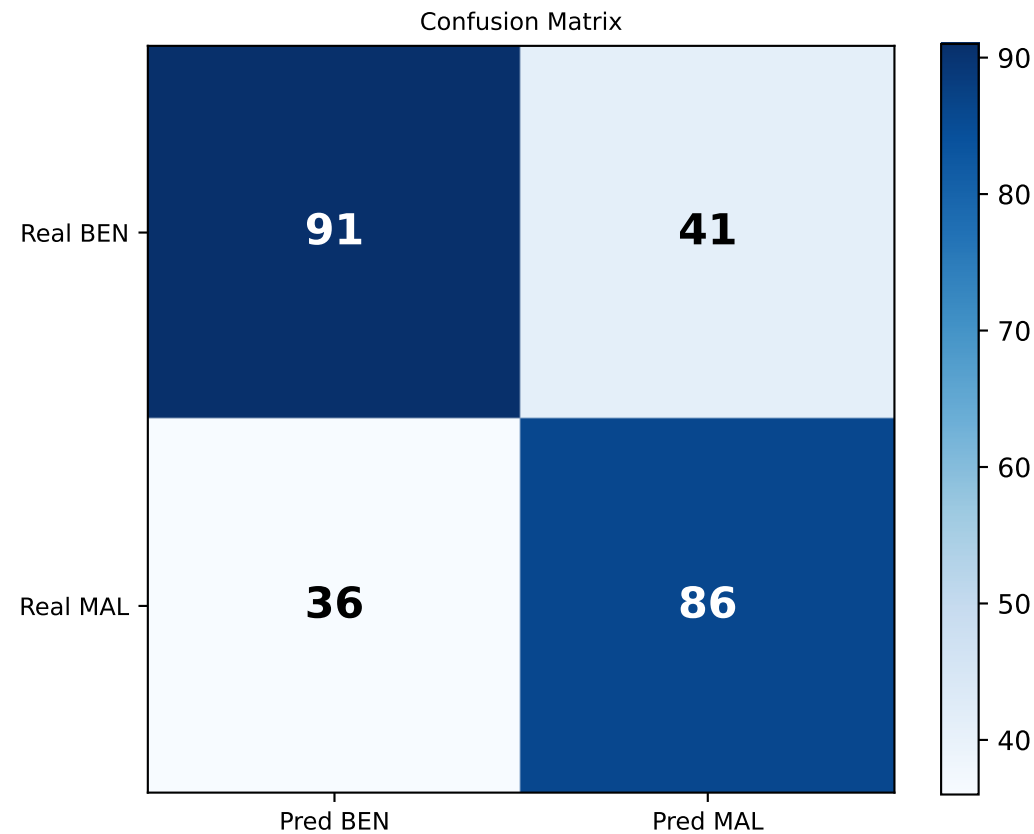
Umbral optimo (max F1 en test) = 0.4010

Accuracy : 0.7165
F1 : 0.7410
Precision : 0.6603
Recall : 0.8443

AUC-ROC : 0.7953
Accuracy : 0.6969 (umbral=0.5)
F1 : 0.6908 (umbral=0.5)
Precision : 0.6772 (umbral=0.5)
Recall : 0.7049 (umbral=0.5)
Score MAL : 0.6683 ± 0.2444
Score BEN : 0.3837 ± 0.2328



fusion_exp01_b2_radiomics — Matriz de Confusión (umbral=0.5)



fusion_exp02_b2_radiomics_selected
EfficientNet-B2 + PyRadiomics (Fusión Multimodal)

Configuración

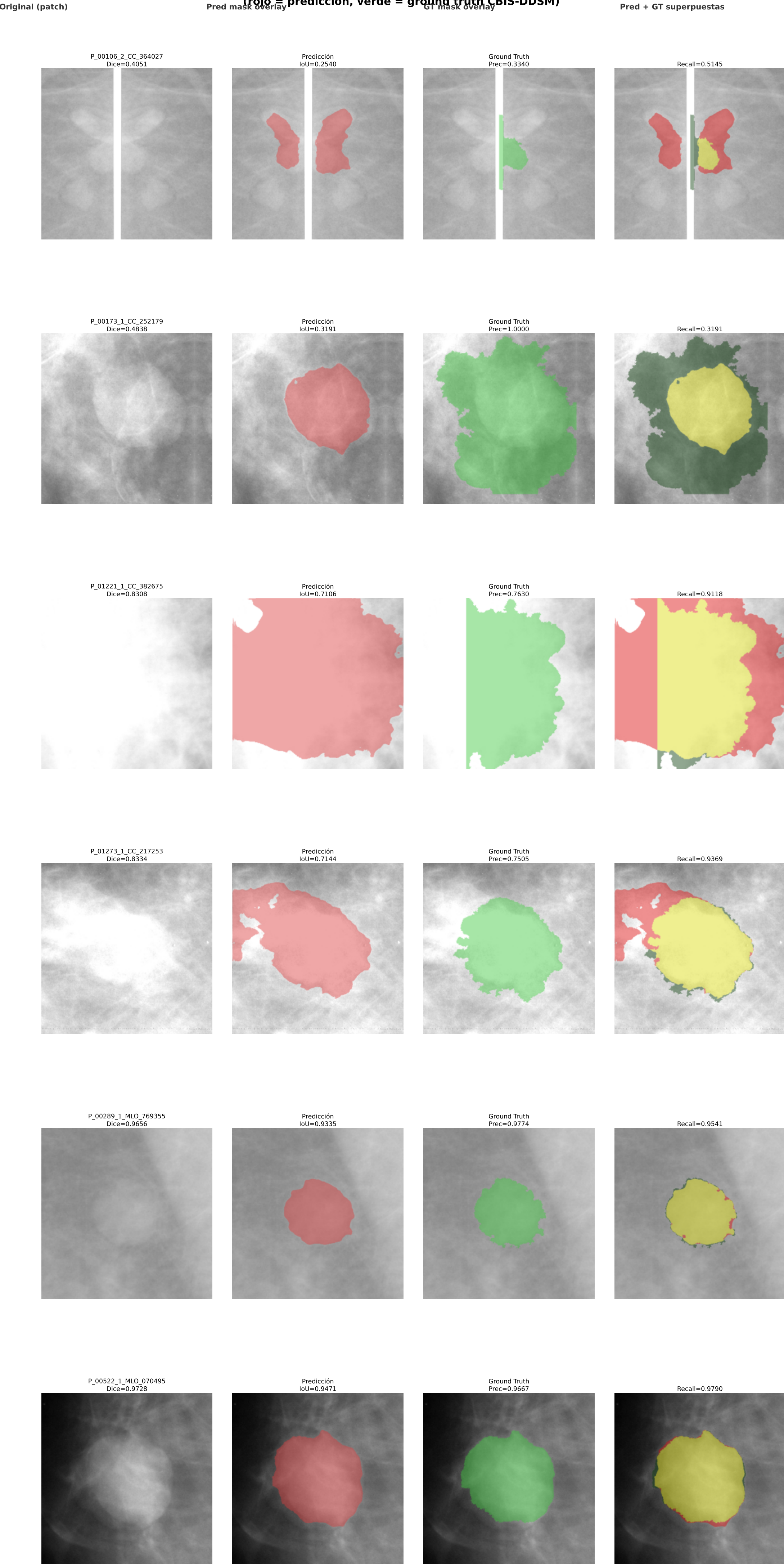
Parámetro	Valor
Backbone	efficientnet_b2
In channels	2
n_radiomics	27
Rad threshold	0.5
Dropout 1	0.4
Dropout 2	0.3
Patch size	512
Batch size	16
Epochs	80
LR backbone	1e-05
LR fusion	0.0001
Weight decay	5e-05
Scheduler	cosine_annealing
Class weights	[1.0, 1.5]
Label smoothing	-
ES metric	val_f1
ES patience	15
Segmentador checkpoint	best.pt

Evaluación test no disponible
(solo checkpoints)

4. Inferencias de Segmentación — Test Set CBIS-DDSM

Mejores / medianos / peores casos por Dice • Comparación predicción vs. ground truth

masses_exp07_patch512_resnet34 — Inferencias en Test Set
(rojo = predicción, verde = ground truth, amarillo = superpuestas)



Predicción (solo) Ground Truth (solo) Intersección (TP)

